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MATH 327 Categorical Data Analysis

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# Final Project Proposal

We used a data set from Kaggle named "Education & Career Success" that provides a view into how students' educational achievements provide "real-world career outcomes." This data set includes 400 records from students' academic performance, their extracurriculars, and their outcomes once they started their career. There are 19 different variables, but we only selected four variables to study due to their expected association.

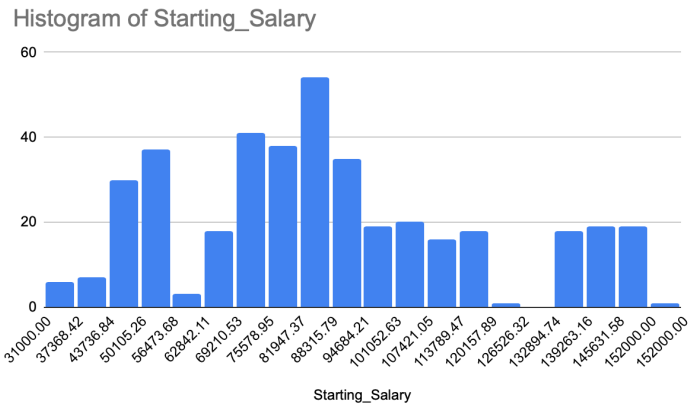
The data set includes a wide range of variables, such as student ID, age, gender, high school GPA, SAT score, university GPA, field of study, internships completed, projects completed, certifications, soft skills score, networking score, job offers, starting salary, career satisfaction, years to promotion, current job level, work life balance, and entrepreneurship. Age, high school GPA, SAT score, University GPA, internships completed, projected completed, certifications, soft skills score, networking score, job offers, starting salary, career satisfaction, years to promotion, and work life balance are all numerical variables. Student ID, gender, field of study, current job level, and entrepreneurship are all categorical variables.

We decided to analyze starting salary, jobs offered, internships completed, and field of study. For internships completed, we decided to group them as 0 internships, 1 internship, 2 internships, and 3 or more internships to list it as a categorical variable. Additionally, we used the same process for jobs offered, grouping them as: 0 jobs offered, 1 job offered, 2 jobs offered,

3 or more jobs offered. For this study, we will be using a combination of these variables to test and analyze how academic and practical experience affect the starting salary for a desired job.

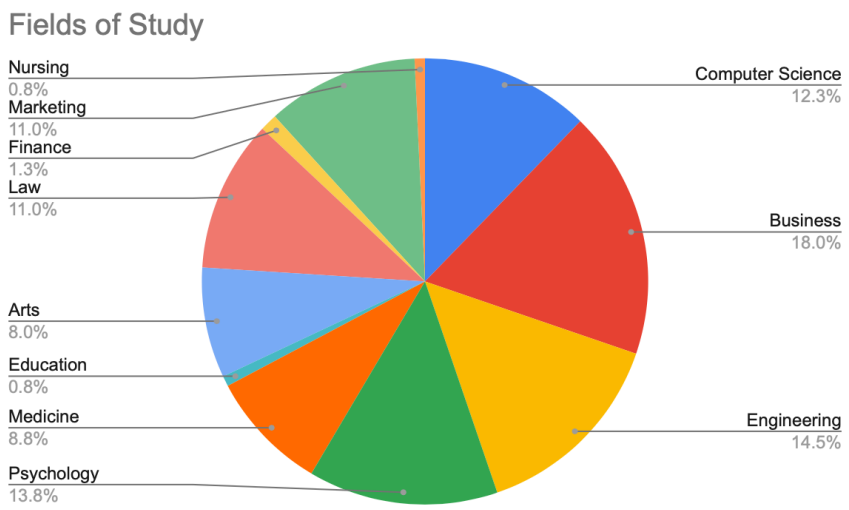
Below are descriptions and visual graphs to understand the variables we are including in this project.

Starting Salary - In this study, starting salary is one of our quantitative variables and it shows how much money each individual makes annually in their first career job. Starting salary is a dependent variable and we are asking ourselves if past college experience and internships have a relationship with starting salary. Starting salary is an important factor in this study and overall life because it is the main way for individuals to provide the necessities for themselves and others that they may be providing for. A histogram can be used to show a clear visualization of how the starting salary values are depicted throughout the study. We can see that the lowest starting salary is $31k and the highest is $152k and the overall dataset is skewed slightly to the right. The distribution has a mean of $87,563 with the most common starting salary being between $81,947 and $88,315.



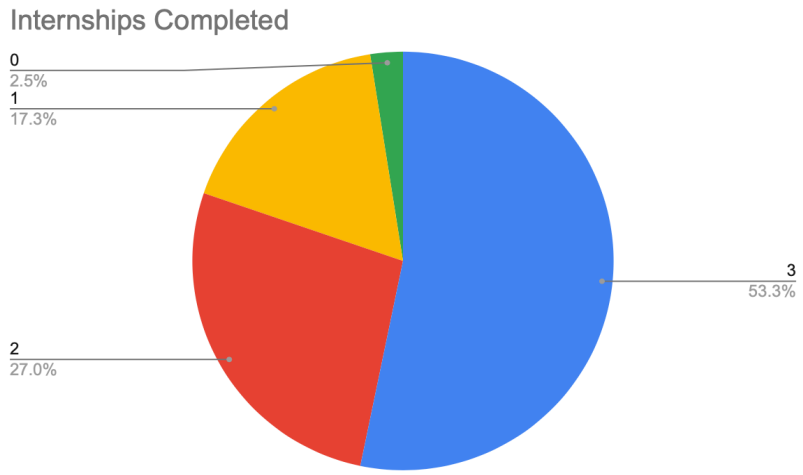

Field of Study - In this dataset, field of study is one of our categorical variables and it shows what each individual graduated with after their college education. Field of Study is a

nominal variable in this dataset based on the variety of majors individuals have in this dataset. Field of Study is an independent variable in this dataset and is a key part in determining the starting salary for the people in this study. Field of study is an important factor because it is a gateway to what one might want to do throughout their life. We can use a Pie Chart to get a look at how many people participated in a field of study. We can see that the most common field of study is business with 18% of the individuals and nursing and education have the least common with 0.8% of the individuals.

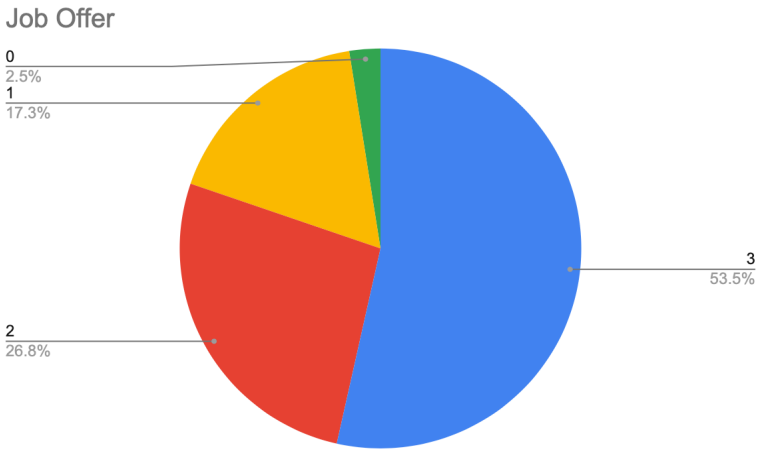



<u>Internships Completed</u> - In this study, internships completed is one of our categorical variables and it shows which individuals completed one (1), two (2), multiple (3), or zero (0) internships. Internships completed is the people in the study completed an Ordinal variable depending on how many internships. This variable is an independent variable in helping determine the future starting salary and whether the internships actually help in determining that annual salary. Internships are an important factor because they help guide someone into the career life and workforce through steps whether it's paid or volunteering. We can visualize this variable using a pie chart to see the distribution of those who have completed an internship,

multiple internships, or never completed an internship. We can see that only 2.5% of individuals never completed an internship while over 50% of people in the study completed more than 2 internships.




Jobs Offered - The data set describes the Job Offers variable as the "total number of distinct full-time job offers received within six months of graduation." This variable is listed from 0 job offers to 5 job offers. We decided to group some of these numbers together instead to test if the amount of job offers altered your future starting salary. We classified these numbers as 0 job offers, 1 job offer, 2 job offers, and three or more job offers. The job offers variable is an ordinal variable since it is a natural listed order. Since we are using this to test the starting salary as a predictor variable, we can conclude that this is independent. The amount of job offers is an important variable to include in the predictors because it usually indicates that the applicant or candidate has optimal skills for the specific career they are trying to achieve.




This project aims to understand how academic and career-related factors influence the annual base salary in USD for the first full-time position through the following research questions.

1. Is there an association between internships completed and job offers? The goal of this question is to determine if the number of internships completed has a statistically significant association with the amount of job offers. A two-way contingency table will be created with the number of internships completed as the rows and the number of job offers as the columns. A chi-square test of independence will then be run with the categorical dependent variable being job offers and the categorical independent variable being the number of internships completed. This test will show us if the distribution of job offers differs significantly across the number of internships. The chi-square statistic, degrees of freedom, and corresponding p-value will be used to evaluate significance. If the result is significant, it would indicate that the number of internships completed is associated with the number of job offers received.
2. Is the proportion of high career satisfaction graduates significantly different between STEM and non-STEM fields? The goal of this question is to determine whether field of

study (STEM vs. non-STEM) is associated with differences in self-reported high career satisfaction. To do this, the independent variable will be field of study (grouped as STEM vs. non-STEM) and the dependent variable as career satisfaction (grouped as high vs. not high). Then, a 2×2 contingency table will be created to summarize the counts of graduates in each category. Proportions of graduates with high career satisfaction will then be computed for each field. Finally, to test whether these proportions are significantly different, a two-proportion z-test will be conducted.

3. Among field of study, internships, and job offers, which loglinear model (testing mutual independence, joint independence, marginal independence, conditional independence, and homogeneous association) best explains their interrelationships?

   The goal of this question is to determine which loglinear model best explains the relationships among fields of study, number of internships and number of job offers. To test this, we will use loglinear analysis in R. Several models including mutual independence, joint independence, marginal independence, conditional independence and homogeneous association will be fitted. The chi-squared statistics and p-values for each model will be compared to see which model fits the data best.

Education & Career Success data set link:

https://www.kaggle.com/datasets/adilshamim8/education-and-career-success